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From Signal to Safeguard: A Capability-Preserving Architecture for Fused Haptic-Biosensor Harassment Filtering

Architectural Foundations: Capability Preservation and Monotone Escalation

The foundational principle governing the design of the fused multimodal haptic-biosensor filter is capability preservation, which manifests as a monotone escalation model. This architecture is not an optional feature but the central tenet that dictates every other aspect of its operation, ensuring that anti-abuse mechanisms cannot become vectors of harm themselves . The core idea is to construct a system state lattice where user capabilities—such as BCI motor control, XR overlays, neural-to-text conversion, and haptic channels—are strictly non-decreasing over time . Any governance action, whether it involves logging, review, or intervention, is implemented by adding new functions or constraints rather than by removing existing ones. Formally, if s_t represents the system state at time t with a set of user capabilities $c(s_t)$, then any subsequent state s_{t+1} must satisfy the condition $c(s_{t+1}) \supseteq c(s_t)$. This invariant guarantees that even under severe harassment or exploit attempts, the user's fundamental autonomy, motor function, and access to support systems remain intact .

This capability-preserving philosophy translates into a three-mode operational policy that scales governance in response to detected threats without compromising user functionality. The first mode is **Normal Mode**, representing baseline operation where the system performs standard tasks without heightened scrutiny . This mode corresponds to a low rogue score, indicating minimal evidence of harassment. The second mode is **Augmented-Logging Mode**, which is activated when the aggregated rogue score exceeds a lower threshold, τ_1 . In this state, the system adds more detailed telemetry and audit trails to its governance set (G), but it does not alter the user's underlying capabilities (C) . This provides enhanced observability and forensic data for later analysis without interrupting the user experience. The third and most restrictive mode is **Augmented-Review Mode**, triggered when the rogue score surpasses a higher threshold, τ_2 . Critically, this mode does not involve outright blocking or disabling actions. Instead, it wraps high-risk commands in a composite action, such as `require_review` ∘

`original_action` . This means that any command capable of affecting sensitive outputs, like haptic feedback or node-interpreter graphs, requires explicit human or multi-signature approval before it can execute . The user's action menu and base capabilities remain fully accessible in all modes, ensuring that the system's defensive posture enhances safety without infringing on sovereignty . This approach directly aligns with the principles of emerging neurorights frameworks, which prioritize mental integrity, cognitive liberty, and protection from algorithmic bias [51] [71] . It also resonates with regulatory requirements like those in the EU AI Act, which mandate comprehensive event logging for high-risk systems and establish clear accountability chains for incident reporting [1] .

The practical implementation of this model relies on a formal separation of user-side capabilities and governance-side functions within the system's state representation . When a potential harassment event is detected, the system's logic evaluates whether a proposed transition would violate the monotonicity invariant. If an action would reduce any part of the user's capability set C, it is disallowed by construction . This prevents scenarios where a user might be locked out of a therapeutic VR environment due to an automated harassment flag raised by another user, or have their haptic feedback disabled during a critical task. Instead, the system constrains the *source* of the threat by wrapping its actions in additional checks. For example, if a node-interpreter graph interaction triggers a high NHSP score, the system doesn't prevent the user from interacting with that node; it simply requires that any command originating from that node must pass through a human reviewer or a multi-sig safety policy . This ensures that the system's response is targeted and proportionate, preserving the user's ability to engage with potentially hostile agents while protecting them from being weaponized by those agents. The audit-and-review workflow becomes the primary mechanism for managing high-risk situations, creating a verifiable record of escalations and decisions that can be examined by the user or regulators to ensure compliance with neurorights [59] . This evidentiary approach, enabled by automatic event recording as required by regulations like the EU AI Act, is crucial for maintaining trust and accountability in a system that interfaces with the brain and body [1] [60] .

The Detection Engine: Hierarchical Risk Scoring and Fused Embeddings

The detection engine of the haptic-biosensor filter is engineered as a sophisticated, multi-layered risk assessment system designed to move beyond simplistic keyword filters and detect nuanced, cross-channel abuse patterns. Its core function is to compute a quantitative "rogue score," denoted as $R(M)$, which represents the cumulative evidence of harassment over a window of interactions . This score is calculated using a weighted sum of risk contributions from four defined harassment families: Neural-Harassment-Spike-Pattern (NHSP), Haptic-Targeting-Abuse (HTA), Prolonged-Session-Abuse (PSA), and Node-Interpreter-Harassment (NIH) . While the architecture is designed to support all four families from inception, they are treated with a hierarchical priority structure reflecting their potential impact on user well-being . HTA and NHSP are assigned the highest priority because they directly target sensory and mental integrity, warranting immediate real-time intervention or redaction when thresholds are exceeded . PSA and NIH, which encompass slower-burn or structural abuses like session manipulation and covert control of interpreter graphs, receive lower priority for immediate blocking but are fully integrated into the rogue score calculation . This ensures that even subtle, long-term harassment campaigns are captured for later audit-and-review workflows, preventing them from falling through the cracks .

The mathematical foundation of the risk scoring model lies in its ability to detect semantic similarity between observed interaction spans and known harassment family templates stored in a unified vector space . For each harassment family $F \in \text{NHSP, HTA, PSA, NIH}$, the system maintains a centroid embedding v_F and a radius r_F . When a new span of interaction with embedding v_s is observed, its proximity to the centroid for each family is calculated. The risk contribution, or weight, of a span s for a given family F is determined by a Gaussian kernel function:

$$\omega_F(s) = \exp(-\alpha_F \, d(v_s, v_F)^2)$$

where $d(v_s, v_F)$ is the distance between the span's embedding and the family's centroid, and \alpha_F > 0 is a scaling parameter that controls the sharpness of the kernel for that family . This formulation allows the system to automatically score new, paraphrased harassment patterns based on their semantic similarity rather than requiring exact textual matches, making the defense adaptive to evolving attack vectors . The final composite rogue score $R(M)$ over a sequence of n spans is then computed as a weighted sum across all families:

$$R(M)=\sum_{j=1}^{n}\sum_{F\in\text{NHSP, HTA, PSA, NIH}}\beta_F\,\omega_F(s_j)$$

where β_F are governance-tuned weights that reflect the prioritization of the harassment families . By assigning higher β_F values to HTA and NHSP, the system amplifies the impact of direct sensory and neural harassment on the total score, triggering escalation faster for these high-priority threats . This fusion of multiple weighted kernels into a single, interpretable score provides a robust metric for driving the monotone escalation policy, transforming abstract concepts of harassment into concrete, calculable indicators of risk .

The effectiveness of this detection engine hinges on the quality of the unified embedding space where both signal-level features and semantic content are represented. This space must be populated with features derived from haptic feedback streams (amplitude, spatial patterns, frequency), biosensor readings (EEG spike rates, heart rate variability, galvanic skin response), AR/VR content (text embeddings, intent), and behavioral interaction logs (repetition rates, session duration) [72]. Research has established strong correlations between these physiological signals and psychological states like stress, anxiety, and emotional arousal, validating their use as reliable inputs for a harassment detection model [5] [73] [79]. For instance, elevated sympathetic nervous system (SNS) activity, often measured via Electrodermal Activity (EDA), is a key indicator of acute stress, while Heart Rate Variability (HRV) reflects autonomic flexibility and resilience [72] [73]. By aligning these diverse feature sets into a single vector space, the system can recognize complex, cross-modal patterns that would be invisible to unimodal detectors. For example, it could identify a sequence where a coercive text prompt (semantic) is consistently paired with stimulation of a sensitive body region (signal) during a prolonged session (behavioral), generating a high rogue score that flags it for review . The continuous refinement of the harassment family centroids v_F through adaptive learning and adversarial red-teaming further enhances the engine's ability to generalize and detect novel forms of abuse .

Data Fusion Layer: Integrating Signal, Semantic, and Behavioral Context

A pivotal innovation of the proposed filter is its data fusion layer, which creates a holistic, multimodal representation by integrating signal-level features, semantic context, and

behavioral patterns into a single, coherent vector space . This fusion is essential for detecting sophisticated, cross-channel abuse that would otherwise be missed by analyzing any single data stream in isolation. The system achieves this by treating neural inputs and outputs as a bidirectional telemetry layer, labeling each interaction with rich contextual information drawn from three distinct domains . The first domain is **signal-level features**, which capture the physical reality of what is happening to the user's body. These features are derived directly from haptic feedback systems and wearable biosensors, including metrics like haptic amplitude, frequency, spatial patterns of stimulation, and a suite of physiological responses such as electroencephalogram (EEG) spike rates, heart rate variability (HRV), galvanic skin response (GSR), respiration rate, and skin temperature [2] [72] [79]. This layer is critical for identifying direct physical targeting (HTA) and psychophysiological distress (NHSP). For example, repeated stimulation of sensitive body regions or sustained haptic patterns that drive stress biomarkers like HRV and GSR beyond safe user-defined envelopes can be reliably flagged as potential abuse [2] [72]. The integration of these signals allows the system to perceive harassment not just as a narrative but as a tangible physiological event [106].

The second domain is **semantic-level features**, which provides the narrative and contextual meaning behind the interactions. This layer analyzes the content of AR/VR environments, dialogue transcripts, and messages exchanged with the node-interpreter graphs . Using techniques from natural language processing (NLP) and transformer-based models, it embeds text and commands into a vector space to understand their semantic meaning, intent, and potential for coercion or manipulation [66]. This allows the system to distinguish between benign requests and harmful narratives, such as attempts at identity-crosslinking, psych-risk manipulation, or the use of AR/VR overlays for stalking or harassment [40]. By maintaining class centroids for harassment families like Node-Interpreter-Harassment (NIH), the system can detect attempts to influence BCI output without explicit consent or inject malicious spike patterns into neuromorphic hardware, regardless of the specific wording used . The fusion of semantic understanding with physiological data is particularly powerful; for instance, the system can correlate a text message containing a coercive command (semantic) with a sudden spike in the user's GSR or a change in EEG patterns (signal), providing stronger evidence of harassment than either signal alone.

The third domain consists of **behavioral-level features**, which analyze the temporal dynamics and patterns of user-agent interactions over time . This involves tracking metrics such as the rate of similar requests, the number of failed policy checks, persistence across sessions, and attempts to bypass established rules . These features are crucial for identifying slow-burn abuse tactics that fall under the PSA and NIH categories.

For example, a bot attempting to erode a user's refusal capability might exhibit a high rate of rapid re-prompting after negative responses, while a power user engaged in legitimate research would show more diverse semantic patterns and fewer evasion signatures . By modeling these behavioral profiles, the system can differentiate between normal heavy use and malicious, persistent engagement . The synthesis of these three layers—signal, semantic, and behavioral—is achieved by aligning their respective feature vectors in time and combining them into a single, high-dimensional vector for each interaction span . This fused representation serves as the input for the risk kernels, enabling the detection engine to recognize complex, multi-faceted abuse patterns. For example, it can identify a conditioning loop where pleasant haptic cues are repeatedly delivered in response to compliant choices and unpleasant cues for refusals, a pattern that would only be apparent when correlating the haptic signal with the semantic content of the prompts and the behavioral history of the user's responses [61]. This deep, integrated analysis is what makes the filter "fully-representable" and capable of supporting the full spectrum of harassment families.

Feature Domain	Key Inputs / Metrics	Relevance to Harassment Families
Signal-Level	Haptic Amplitude/Frequency, Spatial Body Map, EEG Spike Rate, HRV, GSR, Respiration Rate, Skin Temperature [2] [72] [79]	**HTA:** Direct physical targeting of sensitive body regions. **NHSP:** Anomalous neural or physiological stress responses correlated with interactions. **NIH:** Injection of anomalous spike patterns into BCI/haptic systems.
Semantic-Level	Text/Command Embeddings, Dialogue Intent, AR/VR Content, Node-Graph Context [40] [66]	**PSA:** Coercive narratives, repetitive prompts designed to erode refusal. **NIH:** Covert control suggestions, identity-crosslinking attempts, manipulation scripts. **HTA:** Language describing haptic targeting.
Behavioral-Level	Interaction Repetition Rate, Session Duration, Rule-Bypass Attempts, Choice Consistency [37]	**PSA:** Prolonged, consistent, and suspicious actions over time. **HTA/NIH:** Rapid re-prompting or filtering evasion tactics.

Implementation Blueprint: A Modular Rust-Centric Architecture

The technical implementation of the fused multimodal haptic-biosensor filter is guided by a modular, pragmatic, and security-conscious architecture centered around the Rust programming language. This choice leverages Rust's strengths in memory safety and performance, which are critical for a real-time system that processes sensitive biometric

and neural data . The core logic for risk scoring and capability escalation is encapsulated within a dedicated Rust crate named `doctor_labs_superfilter`, ensuring that the most critical computations run in a secure, sandboxed environment . This crate is designed to be composable and reusable, providing a generic framework that can be extended to support various types of harassment detection . The overall system architecture consists of several distinct, decoupled components that work together to provide a complete solution, from data ingestion at the edge to core computation and policy enforcement .

The central component is the Rust crate itself, whose main module, `src/lib.rs`, defines the core data structures and algorithms . This file establishes the fundamental types upon which the entire system is built. It includes an enum `BlacklistFamily` with variants for all known harassment types, including the new additions `NHSP`, `HTA`, `PSA`, and `NIH` alongside legacy families like `CLLN` and `CBCP` . It also defines the `SpanScore` struct, which holds the analytical scores for a single interaction span, including scalar fields for repetition (`y_repetition`), drift (`z_drift`), toxicity (`t_toxicity`), kindness (`k_kindness`), evidentiality (`e_evidentiality`), and a `HashMap` for per-family similarity weights derived from the embedding space . The `RogueScore` struct aggregates these per-span scores into a total rogue score and a breakdown by family, while `RogueConfig` holds the tunable parameters like the alpha and beta weights for the risk kernels and the tau thresholds for escalation . Finally, the `CapabilityMode` enum defines the possible states of the system (`Normal`, `AugmentedLog`, `AugmentedReview`), and the `impl` blocks provide the methods for computing the rogue score and determining the appropriate capability mode .

To bridge the gap between the core Rust logic and the heterogeneous data sources, several adapter modules are employed. The `src/harassment_detector.rs` file acts as a thin wrapper specifically tailored for the NHSP/HTA/PSA/NIH families . It takes raw telemetry from neural I/O streams, biosensors, and session logs, constructs `SpanScore` instances, and delegates the actual scoring and escalation logic to the generic functions in `lib.rs` . This keeps the specialized detection logic separate from the core mathematics, promoting modularity. Another adapter, `src/wordmath.rs`, bridges the system with existing analytical tools by implementing a `AsSpanScore` trait that converts legacy `WordMathAnalysis` results into the standardized `SpanScore` format, ensuring backward compatibility . At the edge of the network, a JavaScript gateway, `doctor-labs-blacklist/src/filters/doctorLabsBlacklist.js`, serves as middleware for web and dashboard clients . This shim is responsible for minimal surface-level pattern matching, calling into the secure Rust crate (likely via FFI or an HTTP interface), handling event logging, and performing the final redaction of sanitized content (e.g., replacing `[HTA-REDACTED]`) . Crucially, this JavaScript layer delegates all

capability decisions—whether to remain in `Normal`, switch to `AugmentedLog`, or enter `AugmentedReview`—back to the trusted Rust backend, preventing any client-side manipulation of system state . This layered, Rust-first architecture provides a robust and secure foundation for the filter, separating concerns cleanly and leveraging the right tool for each job.

Adaptive Defense: Learning from Baselines and Red-Teaming

The filter is designed not as a static blacklist but as a dynamic, living defense system that evolves through two primary mechanisms: adaptive learning from user-specific physiological baselines and proactive adversarial red-teaming. This dual approach ensures the system remains effective against both personalized attacks and novel exploits developed by adversaries. The adaptive learning component focuses on refining the detection engine by tailoring its sensitivity to the unique physiological signature of each user. Since individuals exhibit significant variation in their baseline heart rate, HRV, GSR levels, and neural activity, a one-size-fits-all threshold for "abnormal" is prone to high false positive and false negative rates [72]. To address this, the system will incorporate user-specific baselines to calibrate its internal models and refine its thresholds over time . This process, however, is governed by strict ethical and privacy safeguards. Explicit, granular, and revocable user consent is required for any form of personal data collection or modulation [51]. Furthermore, the system must employ privacy-preserving techniques to avoid exporting raw, identifiable neural or biosensor data . Federated learning architectures, where machine learning models are trained locally on-device without centralizing raw user data, present a promising technical pathway for achieving this goal [3] [29]. Such an approach allows the model to learn generalizable patterns of harassment while respecting the principle of data minimization and user sovereignty [98]. Meta-learning or few-shot adaptation techniques could also be employed to allow the model to quickly adapt to a new user's physiological fingerprint with only a small amount of calibration data, minimizing the burden on the user and enhancing subject independence [72].

The second pillar of the adaptive defense strategy is adversarial red-teaming, a continuous cycle of simulated attacks designed to proactively discover vulnerabilities before they can be exploited by malicious actors. This process involves researchers actively generating new harassment attempts against the system in controlled

environments, mimicking advanced threats like neuronal flooding or scanning attacks on neuromorphic hardware, AI-driven phishing swarms, or ghost-access tactics exploiting Bluetooth vulnerabilities . Every successful bypass or exploit uncovered during these exercises is treated as a valuable data point. This newly discovered attack vector is then systematically converted into a new, formally documented rule for the filter, added as a new test case to the monotone capability lattice, and used to create a new labeled pattern in the harassment embedding space . For example, a successful FLO (neuronal flooding) or SCA (side-channel attack) style exploit on Loihi2-style hardware would result in a new variant of the HTA or NHSP families being formally defined and added to the system's knowledge base . Each of these new rules and patterns would be cryptographically signed within an `EvidenceBundle`, creating an immutable and auditable record of the threat discovery and mitigation process [60]. This creates a virtuous cycle of improvement, where the system's defenses are constantly being tested and strengthened against the latest threats. This methodology aligns with modern cybersecurity practices and is essential for securing frontier AI systems against novel risks [11]. By closing the loop between attack simulation and defensive enhancement, the filter moves beyond reactive measures to build a resilient, self-improving security posture anchored in empirical evidence and continuous validation.

Policy and Governance: Defining Neurorights-Safe Workflows and Rules

The governance of the haptic-biosensor filter is realized through a combination of neurorights-aligned workflows and a structured, interpretable policy-as-code framework. The core of the system's governance is its capability-preserving escalation logic, which ensures that all actions taken are proportional and respectful of user autonomy . As previously detailed, the system operates in three modes: Normal, Augmented-Logging, and Augmented-Review . The transition between these modes is driven by the calculated rogue score, which quantifies the evidence of harassment from the four families (NHSP, HTA, PSA, NIH) . This workflow is fundamentally designed to protect neurorights, a concept that encompasses fundamental rights aimed at safeguarding mental integrity, cognitive liberty, free will, and mental privacy from adverse effects of neurotechnologies [71]. The system's design directly enforces these rights by preventing any reduction in user capabilities, thereby upholding cognitive liberty. By monitoring for neural-spike anomalies and haptic targeting, it protects mental integrity. And by requiring explicit, revocable consent for data collection and modulation, it upholds mental privacy [51] [71].

To make this governance transparent and auditable, the system supports a YAML-based semantic filter syntax for defining blacklisting rules . This policy language is specifically tailored for BCI/XR contexts and is designed to be expressive and interpretable. Each rule definition includes several key fields that provide a complete context for the policy. A typical rule entry would include a unique `id`, a `category` (e.g., 'HAPTIC-ABUSE'), and a `threat_type` (e.g., 'COVERT-CONTROL') that maps to the harassment families . The `pattern` section defines the conditions that trigger the rule, which can be a combination of semantic, signal, and behavioral criteria. For example, a haptic abuse rule might specify an `embedding_similarity` check against a template of aversive touch, combined with a `behavioral_profile` indicating repeated targeting of a specific body region . The rule's `governance` section contains metadata critical for enforcement, including the `neuroright_violated` (e.g., 'MENTAL_INTEGRITY'), the `legal_basis` for the rule, and an `enforcement_hint` that dictates the system's response, such as `redact`, `log-and-audit`, or `review-escalate` . This structured approach ensures that every policy decision is traceable, justifiable, and aligned with the overarching goal of protecting user rights.

Finally, the system's source-detection and session management mechanisms are also designed to be neuroright-safe, avoiding arbitrary restrictions on legitimate users . Instead of applying global, time-based session caps that could unfairly penalize good-faith users, the system computes a per-source "rogue score" based on aggregated behavioral fundamentals like interaction rate, repetition, and bypass attempts . This score drives graduated governance for that specific source, not for the user as a whole . A low-risk session remains in `Normal` mode, while a high-risk source may trigger `AugmentedReview` for its specific activities, such as hardware control or cross-node actions . Session length limits are framed as neuroright-safe policies driven by risk and consent, not by blind timers . For instance, after a long, high-risk session, the system might issue a soft nudge for re-authentication or require explicit consent renewal, but it would not silently disconnect the user . Hard session limits would only apply if the rogue score remains persistently high, signaling a likely automated or hostile source . All such actions are meticulously logged with an `EvidenceBundle` that specifies the reason for the escalation (e.g., "repeated-prompting" or "psych-risk-manipulation pattern"), providing a clear audit trail that verifies the system's behavior is based on objective harassment variables, not arbitrary control . This comprehensive approach to policy and governance ensures that the filter is not only a technical defense but also a principled instrument of user sovereignty.

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